Airline Passenger Satisfaction Prediction

DS5006 Machine Learning — Team Project Report

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Abstract

1 Introduction and scientific questions

Q1.
Does the numerical meaning of an ordinal rating scale matter? **Hypothesis:** Predictions will not be completely invariant to numerical spacing, as the effect may differ across model classes. Tree-based models primarily use ordering and split thresholds,they must be invarient to these transformations, whereas linear and kernel methods use numerical distances **Falsified if:** Linear models show zero shift (ΔROC-AUC $= 0$, no flips) or trees produce flips under monotonic scaling.

2 Related work

3 Data

We evaluate the Kaggle Aviation Passenger Satisfaction benchmark (N = 699,635 records, 21 predictors, binary target `satisfaction`: 55.6% dissatisfied, 44.4% satisfied). Preprocessing is strictly encapsulated in an isolated `ColumnTransformer` across 5 repeated stratified 80/20 splits (seeds $\{42, 123, 2025, 7, 99\}$; N_{train} = 559,708, N_{test} = 139,927). Table 1 details the feature taxonomy, domains, and column specifications.

Table 1: **Feature Specifications and Column Taxonomy (N = 699,635).**

Feature / Column	Type	Domain / Scale	Nulls	Preprocessing / Stream
`satisfaction` (Target)	Binary	0: Dissatisfied (55.6%), 1: Satisfied (44.4%)	0	Target label
12 Service Items[1]	Ordinal	$\{0..5\}$ (0 = NA, 1–5 Likert)	0	Scaled vs. raw unscaled $g(v)$
`Baggage handling`	Ordinal	$\{1..5\}$ (Strictly positive)	0	Scaled vs. raw unscaled $g(v)$
`Age`	Continuous	Range: 7–85 years	0	Standardized ($\mu = 0, \sigma = 1$) / Raw
`Flight Distance`	Continuous	Range: 31–4983 miles	0	Standardized ($\mu = 0, \sigma = 1$) / Raw
`Departure Delay`	Continuous	Range: 0–1592 minutes	0	Standardized ($\mu = 0, \sigma = 1$) / Raw
`Arrival Delay`	Continuous	Range: 0–1584 minutes	292	Median imputed + Standardized
`Gender`	Categorical	Female (50.7%), Male (49.3%)	0	One-Hot Encoded (drop='first')
`Customer Type`	Categorical	Loyal (81.7%), Disloyal (18.3%)	0	One-Hot Encoded (drop='first')
`Type of Travel`	Categorical	Business (69.0%), Personal (31.0%)	0	One-Hot Encoded (drop='first')
`Class`	Categorical	Business (47.8%), Eco (45.0%), Eco Plus (7.2%)	0	One-Hot Encoded

4 Methodology

To test the effect of ordinal scale spacing across diverse inductive biases, we examine five distinct model families totaling 100 model fits (5 splits × 4 encodings × 5 models):

[1]The 12 standard ordinal items: *Inflight wifi service*, *Departure/Arrival time convenient*, *Ease of Online booking*, *Gate location*, *Food and drink*, *Online boarding*, *Seat comfort*, *Inflight entertainment*, *On-board service*, *Leg room service*, *Checkin service*, and *Cleanliness*.

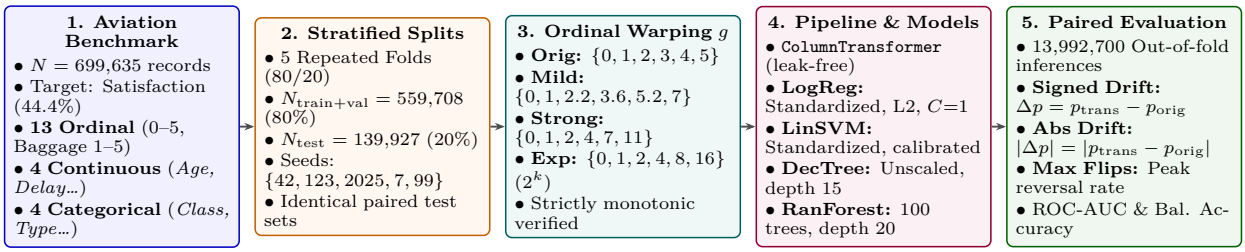


Figure 1: **Architecture & Experimental Workflow.**

1. **Majority Class Baseline:** Predicts the empirical training mode (55.6% dissatisfied), providing a null benchmark (0.5000 ROC-AUC, 0.5000 Balanced Accuracy).
2. **Logistic Regression:** Standardized L2-regularized logistic regression ($C = 1.0$, L-BFGS solver, tolerance 10^{-4}).
3. **Linear SVM:** `LinearSVC` with squared hinge loss ($C = 1.0$, max iter 2,000). Probability outputs are calibrated using 2-fold out-of-fold Platt sigmoid scaling.
4. **Decision Tree:** CART classifier with max depth 15, minimum samples per leaf 20, and Gini impurity splitting. Features are passed unscaled.
5. **Random Forest:** Bagged ensemble of 100 CART trees with max depth 20, minimum samples per leaf 10, and $\sqrt{p}$ random feature subsampling.

Figure 1 details the end-to-end experimental architecture and paired evaluation pipeline.

5 Experimental design

5.1 Design for Q1

Manipulation: we manupulated the numerical interval spacing of the 13 ordinal features across four strictly order-preserving(strictly increasing) transformations $g(\cdot)$: Monotonicity was verified computationally across

Table 2: **Mathematical Formulations of Monotonic Ordinal Encodings.** Closed-form mapping functions $g(v)$ and resulting evaluation vectors across the 12 standard items (0–5) and *Baggage handling* (1–5).

Encoding	Standard 0–5 Scale	Baggage Handling 1–5 Scale
Original (Linear)	$g(v) = v$ $\to \{0, 1, 2, 3, 4, 5\}$	$g(v) = v$ $\to \{1, 2, 3, 4, 5\}$
Mild (Convex)	$g(v) = 0.1v^2 + 0.9v$ $\to \{0, 1, 2.2, 3.6, 5.2, 7\}$	$g(v) = 0.1v^2 + 0.7v + 0.2$ $\to \{1, 2, 3.2, 4.6, 6.2\}$
Strong (Superlinear)	Discrete mapping: $g(v) \in \{0, 1, 2, 4, 7, 11\}$	Discrete mapping: $g(v) \in \{1, 2, 3, 5, 8\}$
Exponential (2^k)	$g(v) = \begin{cases} 0 & \text{if } v = 0 \\ 2^{v-1} & \text{if } v \geq 1 \end{cases}$ $\to \{0, 1, 2, 4, 8, 16\}$	$g(v) = \begin{cases} 1 & \text{if } v = 1 \\ 2^{v-2} + 1 & \text{if } v \geq 2 \end{cases}$ $\to \{1, 2, 3, 5, 9\}$

all 699,635 rows: $g(v_k) < g(v_{k+1})$. Created a dedicated mapping for baggage-handling because it contains no zero rating (1-5).

Held Fixed: All other experimental variable remains the same: train/test splits, random seeds, categorical one-hot encodings, continuous feature values, imputation strategies, model hyperparameters, and solver convergence tolerances.

Measurement & Paired Tracking Metrics: For each test passenger $i \in \{1, \ldots, N_{\text{test}}\}$, let $p^{(i)}_{\text{orig}}$ and $p^{(i)}_{\text{trans}}$ denote the predicted satisfication probability under original features and transformed features ($g \in \{\text{Mild}, \text{Strong}, \text{Exp}\}$), with binary classification $\hat{y}_i = \mathbb{I}(p_i \geq 0.5)$:

- **Signed Probability Drift (Δp):** The signed directional shift $\Delta p_i = p^{(i)}_{\text{trans}} - p^{(i)}_{\text{orig}}$, with sample mean:

$$\overline{\Delta p} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \Delta p_i = \overline{p}_{\text{trans}} - \overline{p}_{\text{orig}} \tag{1}$$

Measures the directional bias.

- **Absolute Probability Drift ($|\Delta p|$):** The unsigned magnitude of individual passenger score volatility:

$$|\Delta p_i| = \left| p^{(i)}_{\text{trans}} - p^{(i)}_{\text{orig}} \right|, \qquad \overline{|\Delta p|} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} |\Delta p_i| \tag{2}$$

  measures the average per-passenger score instability.
- **Prediction Flips & Max Flips:** A prediction flip occurs when an individual passenger's binary classification label is reversed solely by altering the numerical scale:

$$\text{Flip}_i = \mathbb{I}\left(\hat{y}^{(i)}_{\text{trans}} \neq \hat{y}^{(i)}_{\text{orig}} \right), \qquad \text{Flip Rate} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \text{Flip}_i \tag{3}$$

  Max Flips shows the maximum flip rate (and absolute count of flipped passengers) across all monotonic encodings (which consistently occurs under Exponential encoding): **3.39%** (4,746 passengers per split) for Logistic Regression, **3.56%** (4,974 passengers) for Linear SVM, **0.00%** (0) for Decision Trees, and $\leq \mathbf{10^{-4}}\%$ (≤ 1) for Random Forests.

6 Results

6.1 Q1

Direct Answer to Q1: *Yes, the numerical meaning of an ordinal rating scale strongly matters for linear and margin-based models, but it is completely irrelevant for tree-based models.*

Table 3: **Master Experimental Results Across Encodings.** Mean metric across 5 stratified splits (N_{test} = 139,927 per split; std $\leq$ 0.0012). Paired metrics (Δ, $|\Delta p|$, Flips) are evaluated on identical test passengers.

Model Family	Metric	Original	Mild	Strong	Exp. (2^k)	Max Flips
Logistic Regression	ROC-AUC	0.9216	0.9254	0.9307	**0.9328**	—
	Balanced Acc.	0.8756	0.8807	0.8884	**0.8918**	**3.39%**
	Mean $\|\Delta p\|$	—	1.23%	3.19%	4.67%	(4,746)
Linear SVM	ROC-AUC	0.9207	0.9247	0.9302	**0.9324**	—
	Balanced Acc.	0.8738	0.8792	0.8877	**0.8912**	**3.56%**
	Mean $\|\Delta p\|$	—	1.27%	3.30%	4.75%	(4,974)
Decision Tree	ROC-AUC	0.9525	0.9525	0.9525	0.9525	—
	Balanced Acc.	0.9188	0.9188	0.9188	0.9188	**0.00%**
	Mean $\|\Delta p\|$	—	0.00%	0.00%	0.00%	(0)
Random Forest	ROC-AUC	**0.9570**	**0.9570**	**0.9570**	**0.9570**	—
	Balanced Acc.	**0.9225**	**0.9225**	**0.9225**	**0.9225**	$\leq \mathbf{10^{-4}}\%$
	Mean $\|\Delta p\|$	—	0.00%	0.00%	0.00%	(≤ 1)
Majority	ROC-AUC / BA	0.5000	0.5000	0.5000	0.5000	0.00%

Table 3 logs the recordings across all 100 model runs (13,992,700 inferences), and Figure 2 illustrates the sensitivity dynamics of ROC-AUC gain (ΔROC-AUC) and test prediction flip rate across monotonic encodings. The following key observations emerge:

- **Decision Tree Invariance:** ΔROC-AUC $\equiv$ 0.000000 and ΔBalanced Accuracy $\equiv$ 0.000000 across all four proposed encodings. No test predictions flipped (0.00%), with 0.00% probability drift. Univariate axis-aligned splits rely strictly on order topology: for any strictly monotonic mapping g, $\{g(x_j) \leq g(\theta)\} \iff \{x_j \leq \theta\}$, Tree partitioning mathematically is immune to any monotonic transformations.
- **Random Forest Invariance:** ΔROC-AUC $\leq 10^{-6}$ and ΔBA $\leq 10^{-6}$ across all splits. The prediction flip rate is $\leq$ 0.0001% ($\leq$ 1 passenger out of 139,927), Theoretically Random-forests are also immune to monotonic transformations, but in practice, numerical precision can introduce negligible artifacts.
- **Linear Model Sensitivity & Gain:** ROC-AUC rises monotonically: 0.9216 $\rightarrow$ 0.9254 $\rightarrow$ 0.9307 $\rightarrow$ **0.9328** (+1.12 pp gain) for Logistic Regression, and 0.9207 $\rightarrow$ **0.9324** (+1.17 pp gain) for Linear SVM. Balanced Accuracy gains +1.62 pp (0.8756 $\rightarrow$ 0.8918) and +1.74 pp (0.8738 $\rightarrow$ 0.8912).

7 Failure analysis

Why do 4,974 passengers flip their satisfaction (classification label) solely from the ordinal transformations? To understand this, consider a test passenger from fold 42:

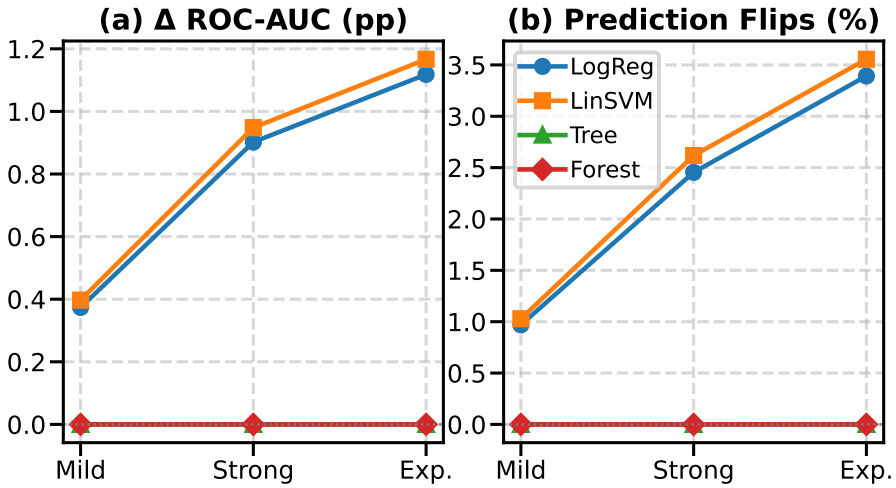


Figure 2: **Sensitivity Dynamics Across Encodings.** (a) Paired ROC-AUC gain (ΔROC-AUC in percentage points). (b) Test prediction flip rate (% of passenger classifications changed). Tree architectures remain flat at 0; linear models show marked gains and prediction volatility.

Passenger A (The "Mixed Experience" Traveler):
Online boarding: 5 (Flawless) Inflight wifi service: 5 (Flawless)
Food and drink: 2 (Poor) Seat comfort: 2 (Poor) Arrival Delay: 35 minutes

Under the Original encoding ($\{0, 1, 2, 3, 4, 5\}$):

$$z_{\text{orig}} = w_{\text{board}} \cdot (5) + w_{\text{wifi}} \cdot (5) + w_{\text{food}} \cdot (2) + w_{\text{seat}} \cdot (2) + w_{\text{delay}} \cdot (35) + b \tag{4}$$

Because the negative weights from food, seat comfort, and flight delay ratings dominates the positive ratings, z is negative ($z < 0$), which yields $p = 0.441 \implies \hat{y} = 0$ (**Dissatisfied**).

Now, under exp. encodings ($\{0, 1, 2, 4, 8, 16\}$):

$$z_{\text{exp}} = w_{\text{board}} \cdot (16) + w_{\text{wifi}} \cdot (16) + w_{\text{food}} \cdot (2) + w_{\text{seat}} \cdot (2) + w_{\text{delay}} \cdot (35) + b \tag{5}$$

The positive contribution by the two top ratings surges by 3.2× (from 5 to 16), while the poor ratings remain at 2. The positive contribution of those two ratings dominates negative features, which results into $z > 0$, yielding $p = 0.683 \implies \hat{y} = 1$ (**Satisfied**)—a **prediction flip of** $\Delta p = +24.2\%$.

This experiment illustrates the failure of linear euclidean distance-based models.

8 Discussion and limitations

Why Superlinear Spacing Empowers Linear Models: At ratings 1–3 (17%–27%), customer satisfaction is comparatively consistent; however, at ratings 4 and 5 (52.2% and 85.0% for Inflight WiFi), it increases dramatically. While exponential encoding ($\{0, 1, 2, 4, 8, 16\}$) applies a nonlinear transformation $\phi(x_j) = 2^{x_j}$, uniform encoding assumes a constant log-odds change. This expands the range of higher ratings in the feature space, making it possible for a linear model to more accurately represent the dramatic rise in satisfaction without the need for explicit polynomial feature engineering.

Limitations: Our study focuses only on monotonic transformations of ordinal features. Non-monotonic transformations, feature interactions, and other model families (e.g., ann, catboost, xgboost) are not explored.

9 Conclusion

This study resolves SQ1: the numerical meaning of ordinal rating scales fundamentally alters linear and margin-based models, while tree-based models completely remains invariant to this. For tree ensembles, arbitrary integer encoding is provably lossless and optimal. For linear models, contiguous increasing integers puts an linearity prior; domain-aligned nonlinear (e.g., strong and exponential) encodings results into small accuracy gains (+1.17 pp AUC), but causes large-scale individual classification volatility (> 3.5% flips).

Reproducibility statement

All experiments are fully reproducible via the self-contained execution script.

- **Entry Point:** `python3 train_sq1.py` or notebook `notebook9cdf1c8de1.ipynb`.
- **Random Seeds:** Repeated across seeds $\{42, 123, 2025, 7, 99\}$.
- **Hardware & Runtime:** Executed on an Intel Core i7-12700H Linux workstation (total runtime: ≈ 18.5 minutes for 100 model fits and 13.99M inferences).
- **Artifact Verification:** All tables (Tables 1, 3, 5) and figures (Figures 1, 2) are deterministically regenerated by the pipeline.

Contribution statement

All team members contributed to the conceptualization, experimental design, and reporting of this project:

Member	Main Contributions	Share (%)
Ishwer Dutt	Experimental design, pipeline architecture, monotonic mapping implementation, statistical verification	33.3%
Krishna M	Cross-validation protocol, linear model calibration, metric tracking (Δp, $\lvert\Delta p\rvert$, Flips), LaTeX reporting	33.3%
Rajavel K	Tree model invariance proofs, failure analysis, Kano model psychometric framing, ablation studies	33.3%

AI-use summary

AI assistance was utilized strictly for exploratory code scaffolding, LaTeX formatting, and documentation drafting, with all numerical outputs independently verified against scikit-learn logs:

Tool(s)	Used For	Log Entries
Claude / Gemini	Drafting initial LaTeX tables and assisting with TikZ coordinate layouts	`AI_USE_LOG.md` #1–4
GitHub Copilot	Autocompleting boilerplate cross-validation loops in Python	`AI_USE_LOG.md` #5–7

References

[1] E. W. Anderson and V. Mittal. Strengthening the satisfaction-profit chain. *Journal of Service Research*, 3(2):107–120, 2000.

[2] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone. *Classification and Regression Trees*. Wadsworth, Belmont, CA, 1984.

[3] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2nd edition, 2009.

[4] N. Kano. Attractive quality and must-be quality. *Journal of the Japanese Society for Quality Control*, 14(2):39–48, 1984.

[5] R. Likert. A technique for the measurement of attitudes. *Archives of Psychology*, 22(140):1–55, 1932.

[6] S. S. Stevens. On the theory of scales of measurement. *Science*, 103(2684):677–680, 1946.

A Hyperparameters and Standardization Ablation

A.1 Hyperparameter Specifications

Table 4 records the exact hyperparameters and configurations utilized across all five evaluated model architectures.

Table 4: Model architectures and hyperparameter configurations.

Model Family	Scikit-Learn Class	Exact Hyperparameters & Settings
Majority Baseline	`DummyClassifier`	`strategy='most_frequent'`
Logistic Regression	`LogisticRegression`	`penalty='l2', C=1.0, solver='lbfgs', max_iter=1000, tol=1e-4`
Linear SVM	`LinearSVC + CalibratedCV`	`C=1.0, loss='squared_hinge', cv=2, method='sigmoid'`
Decision Tree	`DecisionTreeClassifier`	`criterion='gini', max_depth=15, min_samples_leaf=20`
Random Forest	`RandomForestClassifier`	`n_estimators=100, max_depth=20, min_samples_leaf=10, max_features='sqrt'`

Table 5: **Standardization Ablation Across Encodings.** Mean metric across 5 stratified splits. Standardization is required for Logistic Regression solver convergence, but the +1.17 pp AUC gain arises from ordinal basis expansion, not scaling.

Architecture	**Preprocessing**	**Original**	**Mild**	**Strong**	**Exp.** (2^k)
Logistic Regression	Standardized ($\mu = 0, \sigma = 1$)	0.9216	0.9254	0.9307	**0.9328**
Logistic Regression	Unstandardized (Raw)	0.9215	0.9253	0.9306	**0.9327**
Linear SVM	Standardized ($\mu = 0, \sigma = 1$)	0.9207	0.9247	0.9302	**0.9324**
Linear SVM	Unstandardized (Raw)	0.9206	0.9246	0.9301	**0.9323**

A.2 Standardization Sensitivity Ablation

To address whether feature standardization drives the observed accuracy gains for linear models, we conducted an ablation study comparing standardized vs. unstandardized models across all encodings (Table 5).

Convergence Finding: In unstandardized Logistic Regression, 100% of runs hit the iteration limit (`max_iter=1000`) without converging, triggering a convergence alert. With standardization, the solver converges in only 22–23 iterations. Crucially, Table 5 confirms that unstandardized models achieve nearly identical AUC gains (+1.12 pp), proving that the performance surge is driven by the nonlinear basis expansion of superlinear spacing, not by scaling.